

No. of Functional Features Included in the Solution

Date	7 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Accepts Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita from the user.	Frontend (HTML)	Done	High
2	Input Validation	Validates user-entered values before processing.	Flask Backend	Done	Medium
3	HDI Prediction	Predicts the Human Development Index using the trained Linear Regression model.	ML Model (HDI.pkl)	Done	High
4	HDI Category Classification	Classifies the predicted HDI into Very High, High, Medium, or Low Human Development.	Backend Logic	Done	High
5	Result Display	Displays the predicted HDI score and category on the result page.	Result Page	Done	Medium
6	Flask Integration	Connects the frontend with the Machine Learning model.	Flask Application	Done	High
7	Model Loading	Loads the trained HDI model using Pickle during application startup.	Backend	Done	Medium

8	Responsive Web Interface	Provides a simple and user-friendly interface for prediction.	HTML/CSS	Done	Medium
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Result Page
2	Backend / Logic	3	Prediction Logic, Category Classification, Input Validation
3	Database / Storage	1	HDI Dataset (CSV) and HDI.pkl Model
4	API / Integration	1	Flask Route Integration (/predict)
5	Security / Authentication	1	Basic Input Validation